

**ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)**  
**ORGANISATION OF ISLAMIC COOPERATION (OIC)**  
**Department of Computer Science and Engineering (CSE)**

QUIZ-4

DURATION: 30 MINUTES

WINTER SEMESTER, 2022-2023

FULL MARKS: 15

**Math 4543: Numerical Methods**

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

- |                                                                                                                                                             |                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 1. State the fundamental theorems of calculus. Applying these theorems, explain how Euler's method is used to approximate the value of a definite integral. | 3 + 2<br>(CO4)<br>(PO1) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|

**Solution:**

The fundamental theorems of calculus are –

- i. Let  $f$  be a continuous real-valued function defined on a closed interval  $[a, b]$ . Let  $F$  be the function defined, for all  $x$  in  $[a, b]$  by

$$F(x) = \int_a^x f(t)dt$$

Then  $F$  is uniformly continuous on  $[a, b]$  and differentiable on the open interval  $(a, b)$ , and  $F'(x) = f(x)$  for all  $x$  in  $(a, b)$  so  $F$  is an antiderivative of  $f$ .

- ii. If  $f$  is Riemann integrable on  $[a, b]$  then

$$\int_a^b f(x)dx = F(b) - F(a)$$

The idea is to rewrite the integral as the solution of an ODE,

$$\frac{dy}{dx} = f(x), y(a) = 0, \text{ using the 1st Fundamental theorem}$$

where then  $y(b)$  gives us the value of the integral  $\int_a^b f(x)dx$ , according to the 2nd Fundamental theorem. We can use Euler's method to approximate the value of the dependent variable  $y$  at the discrete point  $x = b$ .

**Rubric:**

- 1.5 points each correct fundamental theorem.
- 2 points for the correct explanation.

2. Compare and contrast the following numerical algorithms for solving 1st order ODEs — *Euler’s* method, *Runge-Kutta 2nd order* method, and *Runge-Kutta 4th order* method.

2

(CO3)

(PO1)

| <b>Solution:</b>                                                                     |                                                                             |                                                                                         |                                                                             |
|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Any 2 valid comparisons will suffice. Some comparisons can be—                       |                                                                             |                                                                                         |                                                                             |
| Criteria                                                                             | Euler                                                                       | Runge-Kutta 2nd Order                                                                   | Runge-Kutta 4th Order                                                       |
| Formula                                                                              | $y_{i+1} = y_i + f(x_i, y_i)$                                               | $y_{i+1} = y_i + (a_1k_1 + a_2k_2)h, k_1 = f(x_i, y_i), k_2 = f(x_i + ph, y_i + qk_1h)$ | $y_{i+1} = y_i + (a_1k_1 + a_2k_2 + a_3k_3 + a_4k_4)h$                      |
| Terms of the Taylor series                                                           | First 2 terms.                                                              | First 3 terms.                                                                          | First 5 terms.                                                              |
| Order of the Taylor polynomial                                                       | 1st order — which is why it’s also called the Runge-Kutta 1st order method. | 2nd order — which is why it’s called the Runge-Kutta 2nd order method.                  | 4th order — which is why it’s also called the Runge-Kutta 4th order method. |
| Local Truncation Error                                                               | $O(h^2)$                                                                    | $O(h^3)$                                                                                | $O(h^5)$                                                                    |
| Global Truncation Error                                                              | $O(h)$                                                                      | $O(h^2)$                                                                                | $O(h^4)$                                                                    |
| Variant Approaches                                                                   | None.                                                                       | Has 3 different versions – Heun’s method, Midpoint method, and Ralston’s method         | Has 2 different versions – Runge’s approach and Kutta’s approach.           |
| <b>Rubric:</b>                                                                       |                                                                             |                                                                                         |                                                                             |
| <ul style="list-style-type: none"> <li>1 point for each valid comparison.</li> </ul> |                                                                             |                                                                                         |                                                                             |

3. Determine the approximate value of

6

(CO2)

(PO1)
- $$I = \int_8^{30} \left[ 2000 \ln \left( \frac{140000}{140000 - 2100t} \right) - 9.8t \right] dt$$
- using Simpson’s 1/3 rule (with  $n_1 = 6$ ) and Simpson’s 3/8 rule (with  $n_2 = 6$ ).

|                                                                                   |
|-----------------------------------------------------------------------------------|
| <b>Solution:</b>                                                                  |
| <b>Note:</b> The integrand was incorrect in the original question.                |
| <b>Rubric:</b>                                                                    |
| <ul style="list-style-type: none"> <li>6 points for a correct attempt.</li> </ul> |

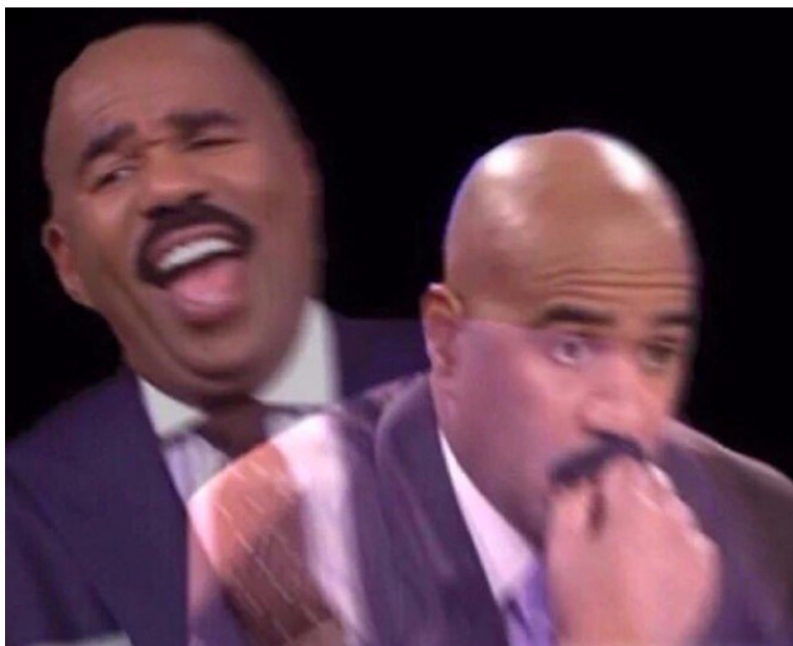
4. An internet meme is defined as a humorous or culturally relevant image, video, or text that spreads rapidly online, often evolving as it is shared and adapted by users. Draw a rough sketch of a math meme related to this course. 2  
(CO69)  
(PO420)

**Warning:** Cringe and lame memes will result in negative marking.

**Solution:**

Me: Last quiz-এ বাচ্চাদের 2 marks ফ্রি দিয়ে দেই।

Also me: \*realizing now I have to give away 8 marks for free\*



**Figure 1:** Steve Harvey.

**Rubric:**

- 2 points for a good meme.